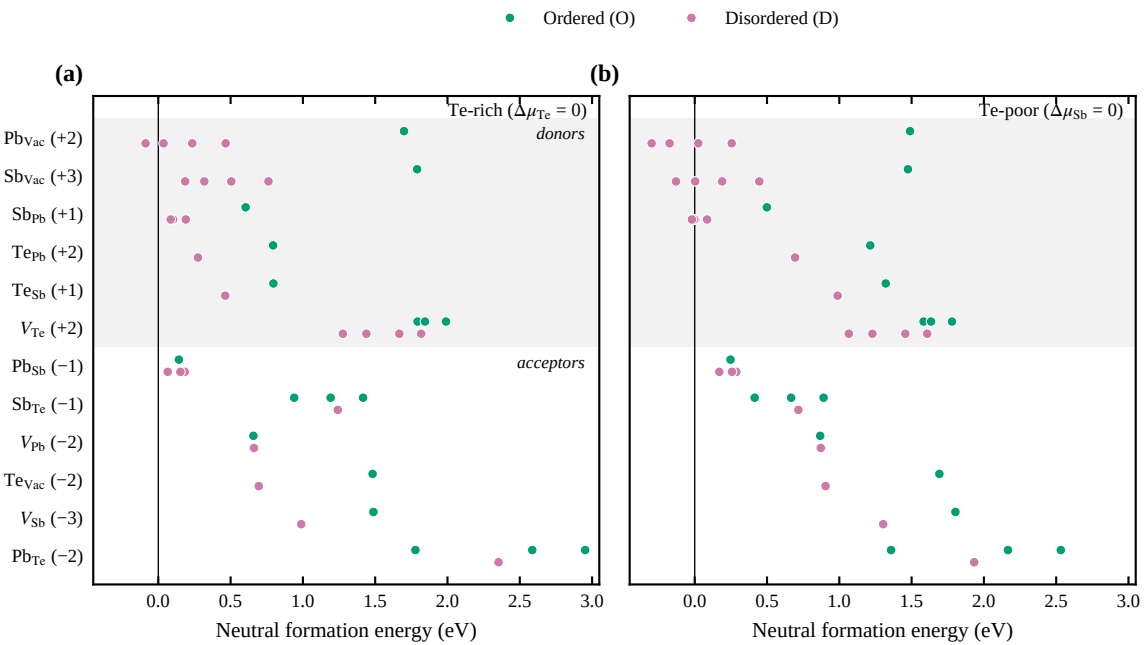
Ordered (O)
Disordered (D)
(a)
(b)
Te-rich ($\Delta\mu_{Te}$ = 0)
Te-poor ($\Delta\mu_{Sb}$ = 0)
donors
acceptors
$Pb_{Vac}$ (+2)
$Sb_{Vac}$ (+3)
$Sb_{Pb}$ (+1)
$Te_{Pb}$ (+2)
$Te_{Sb}$ (+1)
$V_{Te}$ (+2)
$Pb_{Sb}$ (−1)
$Sb_{Te}$ (−1)
$V_{Pb}$ (−2)
$Te_{Vac}$ (−2)
$V_{Sb}$ (−3)
$Pb_{Te}$ (−2)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Neutral formation energy (eV)
Neutral formation energy (eV)